

Assignment 3

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2024-11-01

```
library(R2jags)

## Loading required package: rjags

## Loading required package: coda

## Linked to JAGS 4.3.2

## Loaded modules: basemod,bugs

##
## Attaching package: 'R2jags'

## The following object is masked from 'package:coda':
##
##      traceplot

seed <- 20
```

Example 1

Step 0: Generate some data

```
n <- 100
p <- 0.8
y <- rbinom(n,1,p)
```

```
model <- write("
    model {
      # Prior beliefs about parameter
      p ~ dunif(0,1)

      # Specify likelihood function
      for(i in 1:n){ y[i] ~ dbern(p)    }
    }", "./model.bug")
```

```

# General setup
n.chains <- 4;      n.iter <- 1000
n.burnin <- 100;   n.thin <- 1

# Pass data to JAGS
data_toJAGS <- list("y" = y, "n" = n)

# Specify parameters to keep track of
parameters <- c('p')

# Prepare initial values
myinits <- rep(list(list()), n.chains)
for(i in 1:n.chains){
  myinits[[i]] <- list(p = runif(length(data_toJAGS$n),0,1))
}

```

```

set.seed(seed)

start <- Sys.time()
samples <- jags(data=data_toJAGS,
               parameters.to.save=parameters,
               model="./model.bug",
               n.chains=n.chains, n.iter=1000,
               n.burnin=100, n.thin=n.thin,
               DIC=T, inits=myinits)

```

```
## module glm loaded
```

```

## Compiling model graph
##   Resolving undeclared variables
##   Allocating nodes
## Graph information:
##   Observed stochastic nodes: 100
##   Unobserved stochastic nodes: 1
##   Total graph size: 104
##
## Initializing model

```

```

end <- Sys.time()

cat("Time required to fit this model:", round(end-start, 4), "seconds")

```

```
## Time required to fit this model: 0.0311 seconds
```

```

p <- samples$BUGSoutput$sims.array[,,"p"]

plot(p[,1], type="l")
for(i in 2:n.chains){
  lines(p[,i], col=i)
}

```

